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Karyotypes and Distribution of Tandem Repeat Sequences in *Brassica nigra* Determined by Fluorescence in situ Hybridization

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Brassica nigra · Fluorescence in situ hybridization · Karyotype · (Peri)centromere · Repeat sequence

Abstract
Whole-genome shotgun reads were analyzed to determine the repeat sequence composition in the genome of black mustard, *Brassica nigra* (L.) Koch. The analysis showed that satellite DNA sequences are very abundant in the black mustard genome. The distribution pattern of 7 new tandem repeats (BnSAT13, BnSAT28, BnSAT68, BnSAT76, BnSAT114, BnSAT180, and BnSAT200) on black mustard chromosomes was visualized using fluorescence in situ hybridization (FISH). The FISH signals of BnSAT13 and BnSAT76 provided useful cytogenetic markers; their position and fluorescence intensity allowed for unambiguous identification of all 8 somatic metaphase chromosomes. A karyotype showing the location and fluorescence intensity of these tandem repeat sequences together with the position of rDNAs and centromeric retrotransposons of *Brassica* (CRB) was constructed. The establishment of the FISH-based karyotype in *B. nigra* provides valuable information that can be used in detailed analyses of *B. nigra* accessions and derived allopolyploid *Brassica* species containing the B genome.

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The genus *Brassica* encompasses the 3 widely cultivated diploid species *Brassica rapa* (AA, x = 10), *B. nigra* (BB, x = 8), and *B. oleracea* (CC, x = 9). These species have undergone an additional level of polyploidization through interspecific hybridization, producing 3 cultivated allotetraploid species: *B. napus* (AACC, x = 19), *B. juncea* (AABB, x = 18), and *B. carinata* (BBCC, x = 17) [Nagaharu, 1935].

B. nigra (black mustard) contains a large reservoir of genes conferring various elite characteristics such as resistance to several pathogens [Westman et al., 1999], salt and drought tolerance [Groat, 2003; Pradhan et al., 2011], and oil seed quality [Chèvre et al., 1991; Pradhan et al.,

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2011]. In contrast with the agriculturally important diploid species *B. rapa* and *B. oleracea,* encompassing many vegetable, fodder, and oil crops, our knowledge of genetic information of black mustard is relatively limited, which hindered the transfer of these excellent traits through introgression breeding [Sacristan and Gerdemann, 1986; Sjödin and Glimelius, 1989; Struss et al., 1991; Gerdemann-Kncörck et al., 1994; Wang et al., 2011b]. Additionally, the study of the genome composition and evolution of allopolyploids is restricted due to the lack of information regarding the B genome of *B. nigra*. Recently, the genome sequence of *B. juncea* (AABB) was published, which provides information on the B genome. However, very likely, the B genome in this amphidiploid underwent serious genome rearrangements after allopolyploidization, similar to *B. napus* (AACC) [Yang et al., 2016].

Repetitive elements are predominant components of genomes and can be employed as useful cytological markers to identify species/chromosomes when visualized by FISH and to study phylogenetics [Thompson et al., 1996; Heslop-Harrison, 2000; Meyers et al., 2001]. Several repeats have been characterized in related *Brassica* species including the pericentromeric retrotransposons (such as CRB, pBct, PCRBr, and pBgt), centromeric tandem repeats (CentBrs, TR805, and TR238), and subtelomere-associated satellite repeats (pBrSTR, pBnSTR, and pBoSTR) [Harbinder et al., 1992; Lim et al., 2005, 2007; Koo et al., 2011]. In comparison, information regarding satellite DNA repeats and their distribution patterns in the *Brassica* B genome is lacking.

In this study, we aimed at developing repeat-based cytogenetic markers in order to identify all chromosomes within the black mustard karyotype. Bioinformatics analysis of *B. nigra* whole-genome shotgun reads was employed to characterize major groups of repeats and to select 7 dominant tandem repeats for FISH experiments. An integrated metaphase chromosome karyotype was then established.

Materials and Methods

Plant Materials and Genomic DNA Sequencing

Three *B. nigra* genotypes ("G1/1," "G1/2," and "G1/4") used for cytological studies were kindly provided by Dr. Urich Ryschka, Julius Kühn Institute, Federal Research Center for Cultivated Plants, Germany. Total genomic DNA of "G1/1" was extracted according to Murray and Thompson [1980] and subjected to paired-end Illumina sequencing at The Beijing Genomics Institute (BGI) (3 million read pairs, read length of 90 nucleotides).

Table 1. Repetitive DNA composition of the *B. nigra* genome

Repeat	Genome proportion, %
Retrotransposons	36.29
Ty3/gypsy	26.45
Ty1/copia	7.88
LTR unclassified	1.68
LINE	0.28
DNA transposons	2.01
Helitron	0.10
Satellite	1.59
rDNA	5.27
Unclassified repeats	4.31
Total	49.57

Repeat Composition Analysis

The analysis was performed with the 6 million Illumina paired-end reads, representing 0.86× genome equivalent of *B. nigra* (considering that 1C = 630 Mb; Johnston et al. [2005]). Repeats were identified de novo employing the similarity-based read clustering method as described by Novák et al. [2010] and implemented in the RepeatExplorer pipeline [Novák et al., 2013] using the pipeline default parameters. This analysis resulted in the clustering of overlapping reads, and these clusters represented different families of repetitive sequences. Reads within individual clusters were also assembled to form contigs, representing sequence variants of corresponding repeats. Basic repeat classification was undertaken using a combined approach including the examination of cluster graph shape and similarity searches with DNA and protein databases as implemented in the RepeatExplorer [Novák et al., 2013]. The detection of subrepeats in assembled contigs was performed by similarity dot-plot analysis using a sliding window of 100 bp and similarity cutoff of 40%. Overall repeat composition was calculated excluding clusters of organellar DNA representing contamination of nuclear DNA preparations by chloroplast and mitochondrial DNA.

Repetitive DNAs for FISH Probes

Seven putative satellite repeats (BnSAT13, BnSAT28, BnSAT68, BnSAT76, BnSAT114, BnSAT180, and BnSAT200) were cloned, sequenced, and labeled as FISH probes to detect their distribution over the chromosomes of *B. nigra*. Primer sequences and annealing temperatures of each repeat are listed in online supplementary Table 1 (for all online suppl. material, see www.karger.com/doi/10.1159/000479179). Plasmid inserts containing 45S and 5S rDNA were retrieved from the TIGR Repeat Database (http://www.tigr.org/tdb/e2k1/plant.repeats) and also labeled as FISH probes. Centromeric retrotransposons of *Brassica* (CRB) had been characterized previously (accession number AC166739) [Lim et al., 2005], and the primers (online suppl. Table 1) were designed to amplify 2 different fragments which together covered almost the entire CRB element. The 2 fragments of CRB were mixed to form 1 FISH probe.

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Table 2. Tandem repeats (putative satellites) identified in the *B. nigra* "G1/1" genome

Repeat	Accession number (NCBI, GenBank)	Genome proportion, %	Copy number per 1C	Monomer, bp	Notes
BnSAT13	KX890320	1.025	18,300	352	high similarity to *B. nigra* satellite described by Harbinder et al. [1992]
BnSAT22	KX890321	0.286	5,200	349	up to 80% similarity to BnSAT13
BnSAT28	KX890322	0.145	1,900	485	
BnSAT68	KX890323	0.034	600	354	
BnSAT76	KX890324	0.029	1,300	141	GC-rich (69%), contains palindrome
BnSAT180	KX890326	0.008	720	70	GC-rich, contains palindrome which is in part similar to BnSAT76
BnSAT114	KX890325	0.016	300	332	contains a subrepeat (duplication)
BnSAT195	KX890327	0.007	660	67	GC-rich, palindrome
BnSAT200	KX890328	0.007	370	119	GC-rich

Multicolor FISH Analysis

Multicolor FISH was conducted on mitotic metaphase chromosomes of *B. nigra* according to published protocols [Jiang et al., 1995]. To decompose the cell walls of root tip cells, the root tips were digested at 37°C for 2 h in an enzyme mixture containing 2% cellulase and 1% pectinase (Sigma). The DNA probes were labeled with digoxigenin/biotin-11-dUTP (Roche) or diethylaminocoumarin (DEAC)-5-dUTP (PerkinElmer, Inc.) and detected with anti-digoxigenin antibody coupled with rhodamine (Roche) or anti-avidin antibody conjugated with FITC (Vector Laboratories), while DEAC-labeled probes were detected directly. Chromosomes were counterstained with DAPI in Vectashield antifade solution (Vector Laboratories).

Cytological Measurements and Analysis

Images were captured digitally using a SenSys CCD camera (QIMAGING, RETIGA-SRV, FAST 1394) attached to an Olympus BX61 epifluorescence microscope. Gray-scale images were captured for each color channel and then merged. Measurements were made on the digital images of the FISH signals and chromosomes using Image-Pro Plus 6.0 software (Media Cybernetics), and final image adjustments were done with Adobe Photoshop v6.0. For karyotype construction, chromosomes in 20 metaphase cells were measured.

Quantitative PCR using ChIP DNA

Chromatin immunoprecipitation (ChIP) was performed as before [Liu et al., 2008] using anti-BrCENH3 antibodies. The resulting DNA was used for quantitative real-time PCR (qPCR) as described in Wang et al. [2011a]. The *ACTIN* gene and 5S rDNA (a locus mapped outside of the centromere) were used as negative controls for measuring BrCENH3 binding, and the CRB element previously shown to interact with the kinetochore protein [Wang et al., 2011a] was used as positive control. Primers employed for qPCR are listed in online supplementary Table 1. The difference in the relative fold enrichment (RFE) between the target sequences and *ACTIN* was determined using a *t* test with a significance level of $\alpha = 0.05$.

Results and Discussion

Composition of the Repeat Sequences in the B. nigra *Genome*

Based on a total of 6 million whole-genome shotgun Illumina 90-bp reads, bioinformatics analysis produced thousands of clusters of varying sizes. The clusters making up at least 0.01% of the analyzed reads and thus representing highly to moderately repeated elements [Novák et al., 2010] were analyzed in detail. Combined, these repeats represented 50% of the *B. nigra* genome, and most of them were successfully annotated (Table 1). LTR-retrotransposons were the dominant repeats, with Ty3/gypsy and Ty1/copia representing 26.5 and 7.9% of the genome, respectively. Other repeat types were less abundant, including putative satellite repeats that made up 1.6% of the genome.

Nine clusters were identified that contained putative satellite repeats. A similarity dot plot of their representative contigs is shown in online supplementary Figure 1, and their characteristics are summarized in Table 2. The most abundant satellite (BnSAT13) corresponds to the tandem repeat family previously described by Harbinder et al. [1992] and to the pBnSTR repeat described by Koo et al. [2011]. A sequence variant of BnSAT13 was identified in a different cluster and designated BnSAT22; the 2 repeats are approximately 75–80% similar and likely represent subfamilies of the same satellite. The remaining 7 clusters represent novel tandem repeats. They share no sequence similarity except for BnSAT76, BnSAT180, and to a lesser extent BnSAT195 which contain palindromic sequences with some similarities between these repeats.

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Table 3. Mitotic metaphase chromosomes and relative signal intensity of repeat sequences in *B. nigra* "G1/1"

Chromosome	Relative length[a]	Arm ratio[b]	BnSAT13		BnSAT68		BnSAT28		BnSAT180		BnSAT76		BnSAT114	5S rDNA	45S rDNA	CRB
			S	L	S	L	S	L	S	L	S	L	S	S	S	C
1	14.94	1.37	1[c]	0.83	1	0.92	1.23	0.83	0.38	0.38	–	–	1.21	–	–	4.03
2	14.31	1.43	0.67	–	–	–	0.47	–	0.45	0.36	–	0.34	–	–	–	1.95
3	13.87	1.12	–	0.33	–	–	–	–	0.39	0.38	–	–	–	0.90	–	1.59
4	13.50	1.12	0.35	0.31	0.47	–	1.89	–	0.36	0.32	–	–	–	–	–	1.64
5	12.06	1.49	–	0.25	–	–	0.21	–	0.49	0.09	–	0.56	–	–	0.89	0.64
6	11.20	1.46	–	0.09	0.49	–	0.76	–	0.52	0.31	1.84	0.24	–	–	3.97	2.90
7	10.43	1.39	–	–	–	–	0.46	–	0.08	0.06	–	–	–	–	1.20	1.60
8	9.70	1.13	0.71	0.28	0.47	–	–	0.82	0.46	0.13	0.37	–	–	–	–	0.87

S, short arm; L, long arm; C, centromere.

[a] Relative length = 100 × chromosome length/total complement length.

[b] Arm ratio = length of the long arm/length of the short arm.

[c] The intensity of the BnSAT13 signal on the short arm of chromosome 1 was calibrated as 1, the other signal intensities were converted into relative values.

These 3 repeats as well as BnSAT200 are composed of GC-rich sequences, indicating a possible common origin. Specific primers were designed to generate probes to detect these repeats by FISH with the exception of BnSAT22 and BnSAT195. BnSAT22 was excluded because of its high sequence similarity with BnSAT13, making it difficult to distinguish between these 2 repeats in FISH. BnSAT195 was difficult to clone, probably due to its high GC content.

Distribution Pattern of Tandem Repeat Sequences on B. nigra *Chromosomes*

Figure 1 shows the FISH signals of 6 satellite repeats (BnSAT13, BnSAT180, BnSAT114, BnSAT76, BnSAT28, and BnSAT68) as well as 5S rDNA, 45S rDNA, and CRB on metaphase cells of *B. nigra* root tips. The chromosome lengths and fluorescence intensities of the FISH signals were measured in 20 selected metaphase cells using Image-Pro Plus 6.0 software (Table 3). The relative chromosome lengths varied from 9.70 to 14.94%. All chromosomes are metacentric with an arm ratio of 1.12–1.49. The intensity of the BnSAT13 signal on the short arm of chromosome 1 was calibrated as 1, and the intensities of the other signals were converted into relative values. According to Figure 1 and Table 3, an integrated idiogram of *B. nigra* metaphase chromosomes is shown in Figure 2.

Among the satellite repeats, we found that all somatic chromosomes could be identified when combining FISH signals of BnSAT13 and BnSAT76 (Fig. 1C, C′). The features of each chromosome are as follows:

Chromosome 1. The strongest BnSAT13 signals are located at the ends of both arms.

Chromosome 2. The BnSAT13 signal is located at the end of the short arm, and the BnSAT76 signal is very close to the centromere on the long arm.

Chromosome 3. A BnSAT13 signal is detected at the end of the long arm, and there is no obvious BnSAT76 signal.

Chromosome 4. BnSAT13 signals are located at the ends of both arms.

Chromosome 5. Both signals of BnSAT13 and BnSAT76 are located on the long arm.

Chromosome 6. The BnSAT13 signal is the weakest among the 7 chromosome pairs with BnSAT13 signals, located at the distal end of the long arm. In contrast, the BnSAT76 signal in the short arm is the strongest one detected. The weakest BnSAT76 signal is found very close to the centromere in the long arm.

Chromosome 7. No BnSAT13 and BnSAT76 signals are visible.

Chromosome 8. The BnSAT13 signals are located at the ends of both arms, while the BnSAT76 signal is very close to the centromere on the short arm.

To test the applicability of BnSAT13 and BnSAT76 probes for identifying each chromosome in black mustard genotypes, "G1/2" and "G1/4" were examined. The signals of BnSAT13 and BnSAT76 were found to be consistent with those of the reference in the 2 tested materials (data not shown). Furthermore, although BLAST search showed that all of the 9 satellite repeats with the exception of BnSAT200 had similarities with genome sequences of

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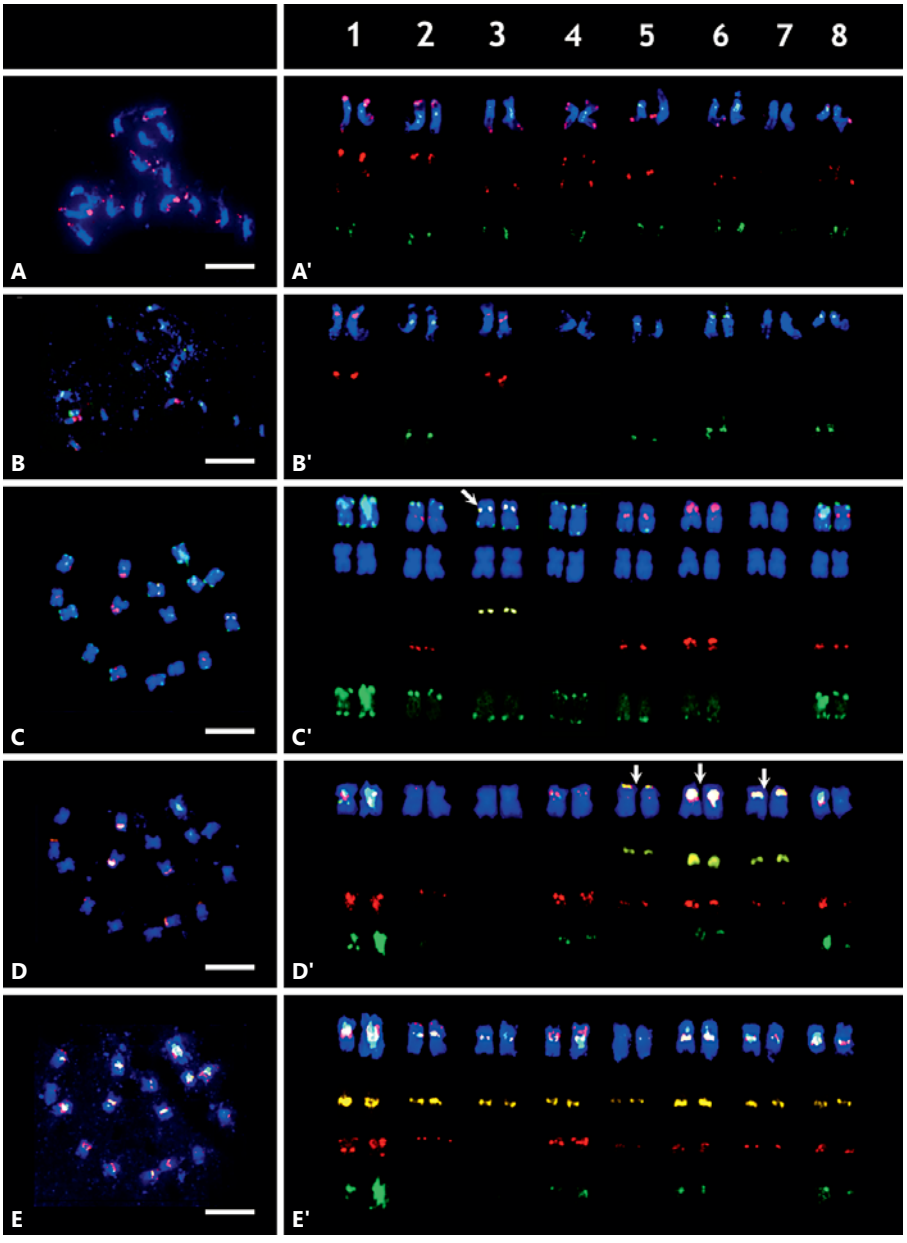


Fig. 1. Metaphase cells (**A–E**) and karyotypes (**A′–E′**) of *Brassica nigra* after FISH with tandem repeats, 5S rDNA, 45S rDNA, and CRB. **A**, **A′** BnSAT13 (red) and BnSAT180 (green). **B**, **B′** BnSAT114 (strong red signal), 5S (weak red signal), and BnSAT76 (green). **C**, **C′** 5S rDNA (yellow, DEAC labeled, indicated by arrow), BnSAT13 (green), and BnSAT76 (red). **D**, **D′** The same cell with 45S rDNA (yellow, DEAC labeled, indicated by arrows), BnSAT28 (red), and BnSAT68 (green). **E**, **E′** CRB (yellow, DEAC labeled), BnSAT28 (red), and BnSAT68 (green). All chromosomes are stained with DAPI (blue). Bars, 5 μm.

B. rapa and *B. oleracea*, no obvious FISH signals of these repeats were detected on *B. rapa* and *B. oleracea* chromosomes (data not shown), indicating that these sequences appear to have evolved in a genome-specific manner with characteristic copy number variation among the A/C and B genomes of *Brassica*.

Numerous cytogenetic karyotypes have been developed for the diploid and allotetraploid *Brassica* species *B. rapa*, *B. oleracea*, and *B. napus* in the Brassicaceae family [Olin-Fatih and Heneen, 1992; Fukui et al., 1998; Lim et al., 2005, 2007; Kim et al., 2009; Xiong and Pires, 2011]. However, to our knowledge, a complete FISH-based karyotype analysis of the B genome of *B. nigra* has not been reported. In this study, we identified satellite repeats and used these to unambiguously identify the 8 chromosomes of *B. nigra*. Our FISH-based karyotyping system will facilitate the characterization of the progeny after distant hybridization, which is a breeding technology for germplasm innovation through introgression breeding. For example, the karyotyping toolkit will allow research-

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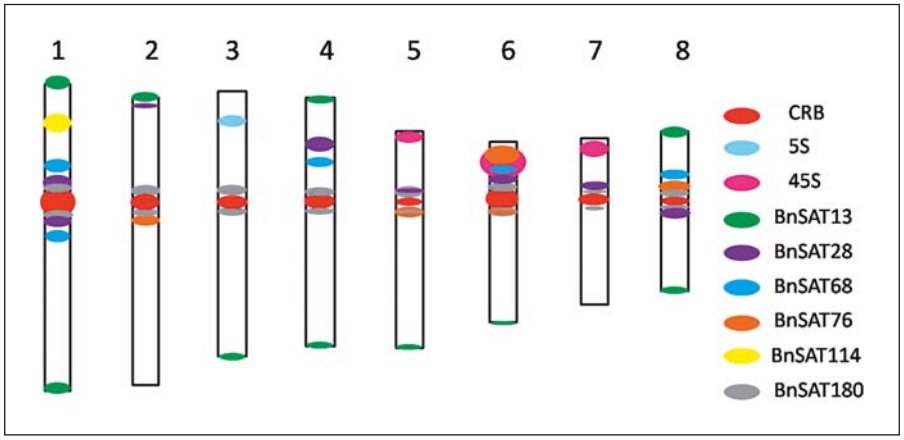


Fig. 2. Idiogram showing the position and intensity of 6 satellite repeats, 45S rDNA, 5S rDNA, and CRB on the 8 *Brassica nigra* chromosomes. The relative length of each chromosome and the arm ratio were drawn based on Table 3. The size of the signal dots represents the relative signal intensity (see Table 3 for details).

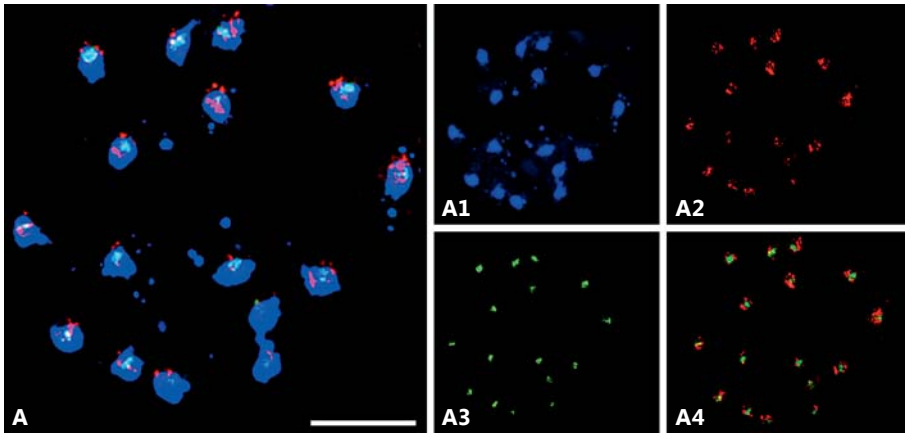


Fig. 3. Distribution of CRB and BnSAT200 sequences in *Brassica nigra*. **A** Merged fluorescence signals of CRB (green) and BnSAT200 (red) on metaphase chromosomes (blue). **A1–A4** The same metaphase showing the chromosomes counterstained with DAPI (**A1**), the red BnSAT200 signals (**A2**), the green CRB signals (**A3**), and the merged BnSAT200 and CRB signals (**A4**). Bar, 10 μm.

ers to correctly identify B genome chromosomes carrying agriculturally significant genes introduced into *Brassica* cultivars and to characterize *Brassica* trisomics, translocation-, addition-, substitution-, and particularly introgression lines in inter-species crosses [Chèvre et al., 1991; Struss et al., 1991; Lysak et al., 2005; Navabi et al., 2010; Wang et al., 2011b]. Additionally, the repeat sequences and karyotyping system of *B. nigra* will facilitate the cytogenetic tracking of homologous chromosomes in allopolyploids and assist in the understanding of chromosome pairing, homologous recombination, and genome evolution in the genus *Brassica* and new resynthesized allopolyploids [Song et al., 1995; Xu et al., 2009; Gaeta and Chris Pires, 2010; Mason et al., 2011].

BnSAT200 Is Located around (Peri)centromeric Regions

Based on our FISH results, BnSAT200 seemed to be located in the (peri)centromeric regions. To test whether this satellite DNA family occupies the (peri)centromere domains, we first conducted dual-color FISH using BnSAT200 and CRB probe cocktails. In mitotic metaphase chromosomes of *B. nigra* (Fig. 3), the green CRB signals are located at the centromeres of all chromosomes, while the red BnSAT200 probes hybridized closely around the green signals with no overlap, while some dots were distant from the CRB signals indicating that BnSAT200 is not centromeric, but occupies the centromere proximal regions. Additionally, we tested by qPCR whether BnSAT200 is enriched in DNA isolated by ChIP with anti-BrCENH3 antibody [Wang et al., 2011a]. Two different regions of the CRB element were used to design primers for real-time PCR analysis. The RFE of the CRB fragments are 9.2 ± 1.4 (mean ± SD) and 9.3 ± 2.6 respectively, indicating that CRB is immunoprecipitated at significantly higher levels than the rDNA (1.5 ± 0.7) control and *ACTIN* (1.0 ± 0.4). In contrast, the RFE of BnSAT200 repeats is 1.8 ± 0.5 which is not significantly different

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from that of the control indicating that the BnSAT200 repeat is not associated with the CENH3 protein in *B. nigra* (online suppl. Fig. 2).

Chromosomal domains such as telomeres and centromeres are favorable sites for satellite occupancy [Peacock et al., 1981]. The satellite repeats CentBr captured centromeric regions of the A and C genomes in *Brassica* species, however no CentBr repeats were detected in the *Brassica* B genome suggesting that centromeric repeats diverged during the evolution of the A/C and B genome lineages. In this study, no centromere-specific satellite repeats were found in *B. nigra*. BnSAT180 and BnSAT200 sequences were located around the pericentromeric regions of *B. nigra*, and these 2 satellites were both GC-rich and might share a common origin.

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Disclosure Statement

The authors have no conflicts of interest to disclose.

References

Chèvre AM, This P, Eber F, Deschamps M, Renard M, et al: Characterization of disomic addition lines *Brassica napus-Brassica nigra* by isozyme, fatty acid, and RFLP markers. Theor Appl Genet 81:43–49 (1991).

Fukui K, Nakayama S, Ohmido N, Yoshiaki H, Yamabe M: Quantitative karyotyping of three diploid *Brassica* species by imaging methods and localization of 45S rDNA loci on the identified chromosomes. Theor Appl Genet 96: 325–330 (1998).

Gaeta RT, Chris Pires J: Homoeologous recombination in allopolyploids: the polyploid ratchet. New Phytol 186:18–28 (2010).

Gerdemann-Kncörck M, Sacristan MD, Braatz C, Schieder O: Utilization of asymmetric somatic hybridization for the transfer of disease resistance from *Brassica nigra* to *Brassica napus*. Plant Breeding 113:106–113 (1994).

Groat CM: The sodium chloride tolerance of *Brassica nigra*. Dissertation, University of California (2003).

Harbinder S, Gupta V, Lakshmikumaran M: Fluidity of the 350 bp tandemly repeated DNA family of *Brassica nigra*. Plant Mol Biol 18: 1213–1216 (1992).

Heslop-Harrison JS: Comparative genome organization in plants: from sequence and markers to chromatin and chromosomes. Plant Cell 12:617–636 (2000).

Jiang J, Gill BS, Wang GL, Ronald PC, Ward DC: Metaphase and interphase fluorescence in situ hybridization mapping of the rice genome with bacterial artificial chromosomes. Proc Natl Acad Sci USA 92:4487–4491 (1995).

Johnston JS, Pepper AE, Hall AE, Chen ZJ, Hodnett G, et al: Evolution of genome size in Brassicaceae. Ann Bot 95:229–235 (2005).

Kim H, Choi SR, Bae J, Hong CP, Lee SY, et al: Sequenced BAC anchored reference genetic map that reconciles the ten individual chromosomes of *Brassica rapa*. BMC Genomics 10: 432–442 (2009).

Koo DH, Hong CP, Batley J, Chung YS, Edwards D, et al: Rapid divergence of repetitive DNAs in *Brassica* relatives. Genomics 97:173–185 (2011).

Lim KB, de Jong H, Yang TJ, Park JY, Kwon SJ, et al: Characterization of rDNAs and tandem repeats in the heterochromatin of *Brassica rapa*. Mol Cells 19:436–444 (2005).

Lim KB, Yang TJ, Hwang YJ, Kim JS, Park JY, et al: Characterization of the centromere and peri-centromere retrotransposons in *Brassica rapa* and their distribution in related *Brassica* species. Plant J 49:173–183 (2007).

Liu Z, Yue W, Li DY, Wang R, Kong XY, et al: Structure and dynamics of retrotransposons at wheat centromeres and pericentromeres. Chromosoma 117:445–456 (2008).

Lysak MA, Koch MA, Pecinka A, Schubert I: Chromosome triplication found across the tribe Brassiceae. Genome Res 15:516–525 (2005).

Mason AS, Nelson MN, Castello MC, Yan G, Cowling WA: Genotypic effects on the frequency of homoeologous and homologous recombination in *Brassica napus* × *B. carinata* hybrids. Theor Appl Genet 122:543–553 (2011).

Meyers BC, Tingey SV, Morgante M: Abundance, distribution, and transcriptional activity of repetitive elements in the maize genome. Genome Res 11:1660–1676 (2001).

Murray MG, Thompson WF: Rapid isolation of high molecular weight plant DNA. Nucleic Acids Res 8:4321–4325 (1980).

Nagaharu U: Genome analysis in *Brassica* with special reference to the experimental formation of *B. napus* and peculiar mode of fertilization. Jpn J Bot 7:389–452 (1935).

Navabi ZK, Parkin IA, Pires JC, Xiong Z, Thiagarajah MR, et al: Introgression of B-genome chromosomes in a doubled haploid population of *Brassica napus* × *B. carinata*. Genome 53:619–629 (2010).

Novák P, Neumann P, Macas J: Graph-based clustering and characterization of repetitive sequences in next-generation sequencing data. BMC Bioinformatics 11:378 (2010).

Novák P, Neumann P, Pech J, Steinhaisl J, Macas J: RepeatExplorer: a Galaxy-based web server for genome-wide characterization of eukaryotic repetitive elements from next generation sequence reads. Bioinformatics 29:792–793 (2013).

Olin-Fatih M, Heneen WK: C-banded karyotypes of *Brassica campestris*, *Brassica oleracea* and *Brassica napus*. Genome 35:583–589 (1992).

Peacock WJ, Dennis ES, Rhoades MM, Pryor AJ: Highly repeated DNA sequence limited to knob heterochromatin in maize. Proc Natl Acad Sci USA 78:4490–4494 (1981).

Pradhan A, Nelson MN, Plummer JA, Cowling WA, Yan G: Characterization of *Brassica nigra* collections using simple sequence repeat markers reveals distinct groups associated with geographical location, and frequent mislabelling of species identity. Genome 54:50–63 (2011).

Sacristan MD, Gerdemann M: Different behavior of *Brassica juncea* and *B. carinata* as sources of *Phoma lingam* resistance in experiments of interspecific transfer to *B. napus*. Plant Breeding 97:304–314 (1986).

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Sjödin C, Glimelius K: Transfer of resistance against *Phoma lingam* to *Brassica napus* by asymmetric somatic hybridization combined with toxin selection. Theor Appl Genet 78: 513–520 (1989).

Song K, Lu P, Osborn C: Rapid genome change in synthetic polyploids of *Brassica* and its implications for polyploidy evolution. Proc Natl Acad Sci USA 92:7719–7723 (1995).

Struss D, Bellin U, Robbelen G: Development of B-genome chromosome addition lines of *B. napus* using different interspecific *Brassica* hybrids. Plant Breeding 106:209–214 (1991).

Thompson HL, Schmidt R, Dean C: Identification and distribution of seven classes of middle-repetitive DNA in the *Arabidopsis thaliana* genome. Nucleic Acids Res 24:3017–3022 (1996).

Wang G, He Q, Liu F, Cheng Z, Talbert PB, Jin W: Characterization of CENH3 proteins and centromere-associated DNA sequences in diploid and allotetraploid *Brassica* species. Chromosoma 120:353–365 (2011a).

Wang G, Tang Y, Yan H, Sheng XG, Hao WW, et al: Production and characterization of interspecific somatic hybrids between *Brassica oleracea* var. *botrytis* and *B. nigra* and their progenies for the selection of advanced prebreeding materials. Plant Cell Rep 30:1811–1821 (2011b).

Westman AL, Kresovich S, Dickson MH: Regional variation in *Brassica nigra* and other weedy crucifers for disease reaction to *Alternaria brassicicola* and *Xanthomonas campestris* pv. *campestris*. Euphytica 106:253–259 (1999).

Xiong Z, Pires JC: Karyotype and identification of all homoeologous chromosomes of allopolyploid *Brassica napus* and its diploid progenitors. Genetics 187:37–49 (2011).

Xu Y, Zhong L, Wu X, Fang X, Wang J: Rapid alterations of gene expression and cytosine methylation in newly synthesized *Brassica napus* allopolyploids. Planta 229:471–483 (2009).

Yang J, Liu D, Wang X, Ji C, Cheng F, et al: The genome sequence of allopolyploid *Brassica juncea* and analysis of differential homoeolog gene expression influencing selection. Nat Genet 48:1225–1232 (2016).

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